

C++ Coding Practice Questions (Functions)

Question 1: Function Prototype and Passing Data by Value

Topic Covered: Function Prototype, Passing Data by Value

Write a C++ program that:

- Declares a function prototype named `calculateSquare()`.
- Accepts an integer as an argument.
- Returns the square of the number.
- In the `main()` function, input a number from the user and display the returned result.

Expected Concepts:

- Function prototype
 - Function definition
 - Passing arguments by value
 - Returning a value
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Question 2: Returning a Boolean Value

Topic Covered: Boolean Functions, Return Statement

Write a C++ program that:

- Creates a Boolean function named `isEven()`.
- Accepts an integer as a parameter.
- Returns `true` if the number is even; otherwise returns `false`.
- In `main()`, ask the user for a number and display:
 - "The number is Even."
 - "The number is Odd."

Expected Concepts:

- `bool` return type
- `return` statement
- Conditional statements

Question 3: Local, Global, and Static Variables

Topic Covered: Local Variables, Global Variables, Static Local Variables

Write a C++ program that:

- Declares a global variable named `totalCalls`.
- Creates a function named `displayCounter()`.
- Inside the function:
 - Declare a **static local variable** to count how many times the function is called.
 - Increment both the static variable and the global variable each time the function executes.
- Call the function five times from `main()` and display both counters after each call.

Expected Concepts:

- Global variables
- Local variables
- Static local variables
- Function calls

Question 4: Menu-Driven Calculator Using Functions

Topic Covered: Menu-Driven Program, Functions, Return Values

Write a menu-driven C++ program that performs the following operations using **separate functions**:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Requirements:

- Display the menu repeatedly until the user selects Exit.
- Each mathematical operation must be implemented in a separate function.
- Use function prototypes.
- Return the calculated result from each function.

Expected Concepts:

- Function prototypes
- Menu-driven programming
- Looping
- Returning values

Question 5: Student Result Analyzer

Topic Covered: Functions, Passing Data by Value, Boolean Functions

Write a C++ program that:

- Inputs marks of **five subjects**.
- Creates separate functions to:
 - Calculate the total marks.
 - Calculate the average marks.
 - Determine whether the student has passed using a Boolean function (`isPass()`), where passing requires an average of **50 or above**.
- Display:
 - Total Marks
 - Average Marks
 - Pass/Fail Status

Expected Concepts:

- Multiple functions
- Passing data by value
- Returning integer and floating-point values
- Returning Boolean values